EE5312: VLSI Technology

Assignment: Virtual Fabrication of a PN Junction Diode and a Self-Aligned n-Channel MOSFET in SEMulator3D

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Overview

Two devices were virtually fabricated in Coventor SEMulator3D (v13.1): a PN junction diode (Part A) and a self-aligned n-channel MOSFET (Part B). Both models use a 200 nm × 200 nm simulation domain, a 100 nm thick p-type (100) silicon wafer and a voxel resolution of 1 nm. For each device, two of the three assigned parameters were kept constant within the group and the third was varied, as assigned to Group 8: the *implantation dose* Q for the PN diode and the *longitudinal range* R_p for the MOSFET. Part A is reported first, followed by Part B, which contains the four items required by the assignment: (1) the MOSFET process flow, (2) the final 3D structure, (3) the cross-section with doping profile and (4) the source/drain junction depth calculation. The values used in this report are $Q = 1 \times 10^{13}\,\mathrm{cm}^{-2}$ (diode) and $R_p = 20\,\mathrm{nm}$ (MOSFET).

Table 1: Simulation parameters.

Parameter	PN diode	MOSFET
Substrate doping (boron), N_B	$1 \times 10^{15}\,\mathrm{cm}^{-3}$	$1 \times 10^{15}\,\mathrm{cm}^{-3}$
Gate oxide thickness	–	20 nm
Implant species	Phosphorus	Arsenic
Implant dose, Q	$1 \times 10^{13}\,\mathrm{cm}^{-2}$ (**varied**)	$1 \times 10^{15}\,\mathrm{cm}^{-2}$
Longitudinal range, R_p	20 nm	20 nm (**varied**)
Longitudinal straggle, ΔR_p	5 nm	5 nm
Radial (lateral) range	5 nm	0 nm
Tilt / twist	0° / 0°	0° / 0°

Group 8 Parameter Plan

The course sheet assigns every group one parameter per device that is varied among its members, while the other two parameters are held at a common value by everyone in the group.

Table 2: Parameters held constant and varied within Group 8.

Device	Parameter	Group 8
PN diode	Substrate doping, N_B	Constant, $1 \times 10^{15}\,\mathrm{cm}^{-3}$
	Implantation dose, Q	**Varied**
	Longitudinal range, R_p	Constant, 20 nm
MOSFET	Implantation dose, Q	Constant, $1 \times 10^{15}\,\mathrm{cm}^{-2}$
	Longitudinal range, R_p	**Varied**
	Gate oxide thickness and N_B	Constant, 20 nm and $1 \times 10^{15}\,\mathrm{cm}^{-3}$

Proposed values. The sheet specifies *which* parameter is varied but not its values, so a set of five values, one per member, was proposed within the group (Table 3). For the diode the dose is stepped by roughly half a decade from 1×10^{12} to $1 \times 10^{14}\,\mathrm{cm}^{-2}$; for the MOSFET the range is stepped in 5 nm increments from 10 to 30 nm. The spacing was chosen so that the effect of the varied parameter on the junction depth is clearly visible across the group. The junction depths listed are calculated with the Gaussian-profile expression used later in this report (straggle 5 nm, $N_B = 1 \times 10^{15}\,\mathrm{cm}^{-3}$).

Table 3: Values proposed for the five members of Group 8, with calculated junction depths. The row in bold is the one simulated in this report.

	PN diode ($R_p = 20$ nm)		MOSFET ($Q = 1 \times 10^{15}\,\mathrm{cm}^{-2}$)	
Member	Q (cm^{-2})	x_j (nm)	R_p (nm)	x_j (nm)
1	1×10^{12}	38.3	10	36.1
2	3×10^{12}	39.7	15	41.1
3 (this report)	1×10^{13}	**41.2**	**20**	**46.1**
4	3×10^{13}	42.5	25	51.1
5	1×10^{14}	43.8	30	56.1

Values chosen for this report. I took the middle value of each sweep: $Q = 1 \times 10^{13}\,\mathrm{cm}^{-2}$ for the PN diode and $R_p = 20$ nm for the MOSFET. Sitting at the centre of both sweeps, these results act as the reference point for the group, with the other members' simulations falling symmetrically on either side. All remaining parameters are the group constants listed in Tables 1 and 2. The fabrication steps for each device follow.

1 Part A: PN Junction Diode

1.1 Layout

The layout (`pn.cat`, top cell `Overall`) contains three concentric square layers (Table 4, Fig. 1).

Table 4: PN diode mask layers (coordinates in nm).

Layer	Rectangle (x_1, y_1)–(x_2, y_2)	Purpose
`layer1`	$(-100, -100)$–$(100, 100)$	Simulation extent
`Doping`	$(-30, -30)$–$(30, 30)$	Implant window
`Metal`	$(-40, -40)$–$(40, 40)$	Top gold contact

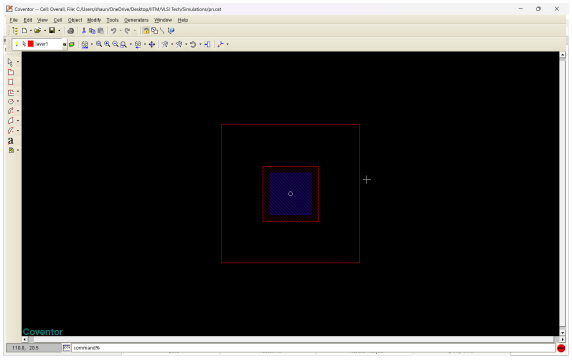


Figure 1: PN diode layout: `layer1` outline, `Metal` (red, hatched) and `Doping` (blue).

1.2 Process flow

The diode follows the 12-step flow from the course slides (Table 5).

Table 5: PN diode process steps.

#	SEMulator3D action	Settings / purpose
1	Wafer Setup	Si_Xtal, (100), 100 nm, boron 1×10^{15} cm^{-3} (p-type substrate)
2	Straight Deposit	SiO_2 (thermal), 20 nm: masking oxide
3	Planarizing Deposit	Photoresist, 50 nm
4	Expose Material	Mask `Doping`, polarity *Dark*: opens the implant window
5	Straight Etch	SiO_2, selective, 20 nm: exposes Si in the window
6	Implant (Analytical)	P, 1×10^{13} cm^{-2}, $R_p = 20$ nm, $\Delta R_p = 5$ nm, radial 5 nm, tilt/twist 0°
7	Straight Deposit	Au, 20 nm, both wafer sides
8	Lift-Off	Removes resist and the Au on it; back-side Au remains as the bottom contact
9	Planarizing Deposit	Photoresist, 50 nm
10	Expose Material	Mask `Metal`, polarity *Dark*, 100 nm
11	Straight Deposit	Au, 20 nm, both wafer sides
12	Lift-Off	Leaves the Au top contact over the n-region

The resist and oxide mask the implant everywhere except the `Doping` window, so phosphorus forms an n-type pocket in the p-type substrate there, giving the PN junction. Gold on the front (over the window) and on the back of the wafer forms the two diode contacts.

1.3 Final 3D structure

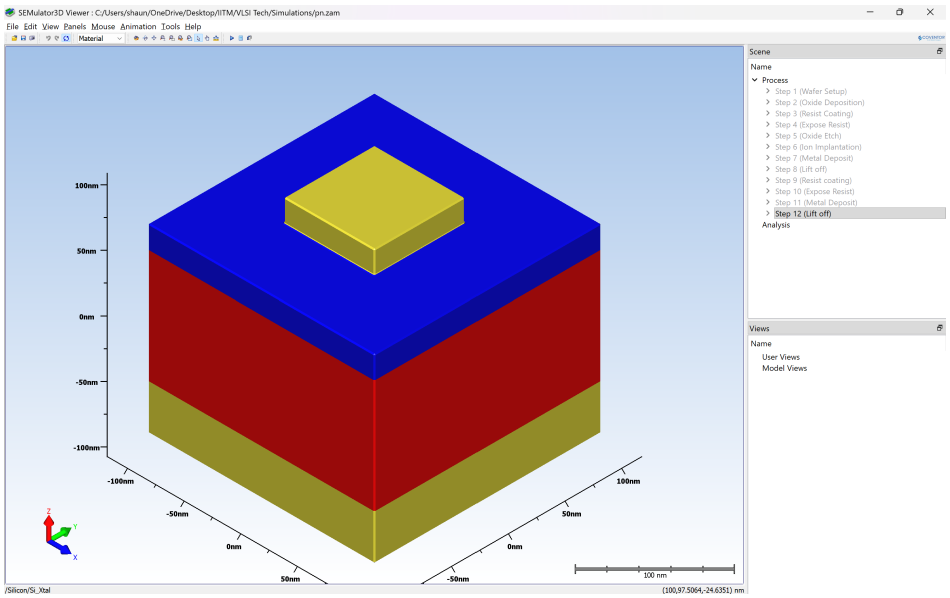


Figure 2: Final PN diode structure after step 12. Red: Si; blue: SiO_2; gold: top contact and back contact.

1.4 Cross-sectional view with doping profile

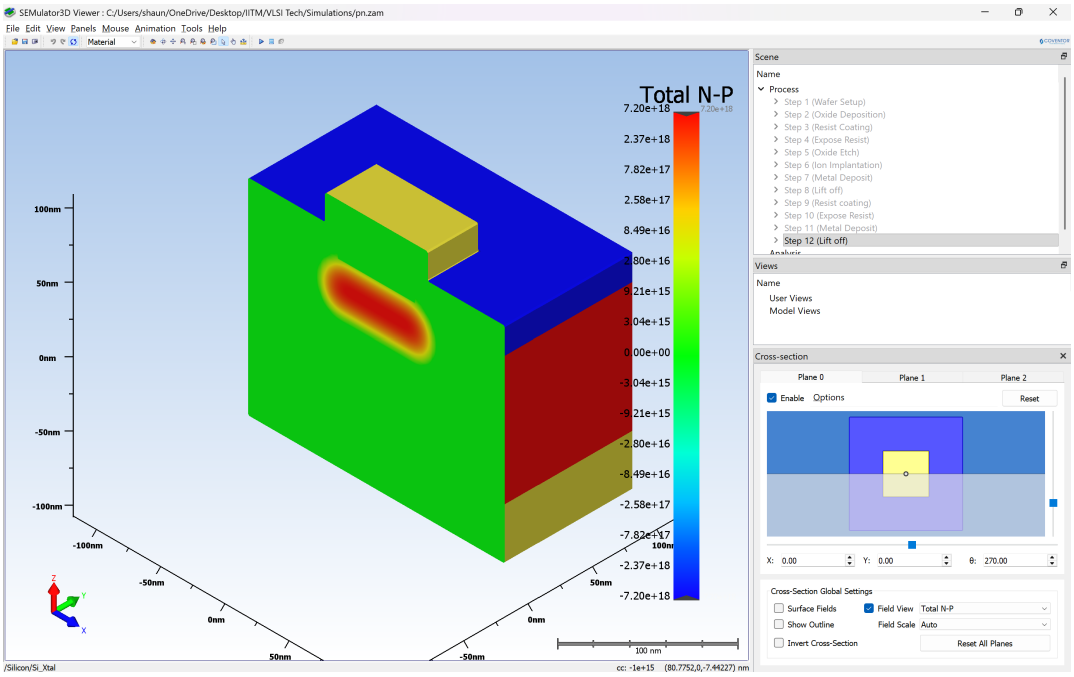


Figure 3: Diode cut open at $y = 0$, coloured by Total N–P. The n-type pocket (red/orange) lies under the implant window. The lightly doped p substrate ($-1 \times 10^{15}\,\mathrm{cm}^{-3}$) appears close to zero (green) on this scale. The resist and oxide stopped the dopant at their surfaces.

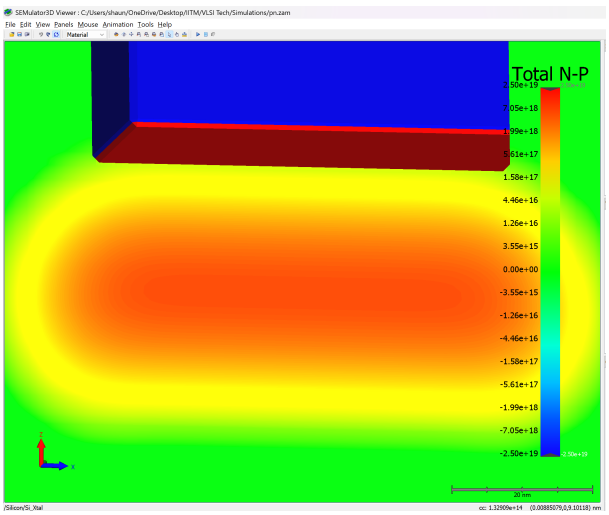
1.5 Junction depth

Using Eq. (2) (derived in Section 2.4) with $Q = 1\times10^{13}\,\mathrm{cm}^{-2}$, $\Delta R_p = 5\,\mathrm{nm}$ and $N_B = 1\times10^{15}\,\mathrm{cm}^{-3}$:

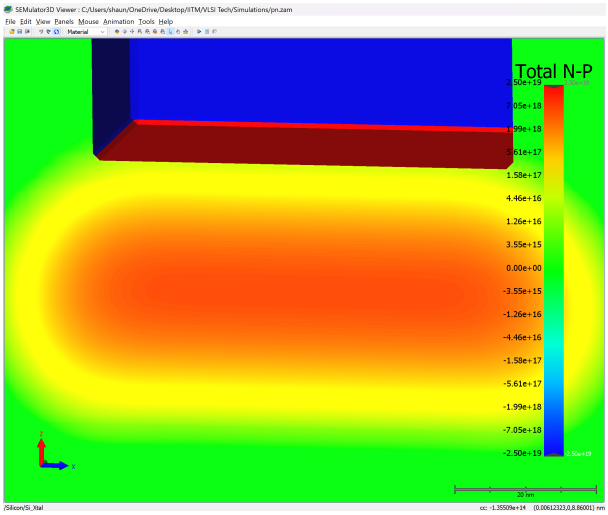
$$N_p = \frac{1 \times 10^{13}}{\sqrt{2\pi} \times 5 \times 10^{-7}} = 7.98 \times 10^{18}\,\mathrm{cm}^{-3}, \qquad \ln\frac{N_p}{N_B} = \ln(7979) = 8.98,$$

$$x_j = 20 + 5\sqrt{2 \times 8.98} = 20 + 21.2 \approx \boxed{41.2\,\mathrm{nm}}.$$

On the simulated cross-section (Fig. 4), the net doping changes sign between $z = 9.10\,\mathrm{nm}$ ($+1.33 \times 10^{14}\,\mathrm{cm}^{-3}$) and $z = 8.86\,\mathrm{nm}$ ($-1.36 \times 10^{14}\,\mathrm{cm}^{-3}$). Linear interpolation puts the junction at $z = 8.98\,\mathrm{nm}$. With the silicon surface at $z = 50\,\mathrm{nm}$, the simulated $x_j = 41.0\,\mathrm{nm}$. This agrees with the calculation to within $0.2\,\mathrm{nm}$ (0.5 %), less than one 1 nm voxel.



(a) Last positive reading, $z = 9.10\,\text{nm}$.



(b) First negative reading, $z = 8.86\,\text{nm}$.

Figure 4: Cursor readings of the net doping (status bar, bottom right) used to locate the diode junction.

Effect of the varied dose. With R_p, ΔR_p and N_B fixed, the dose enters only through the logarithm in Eq. (2), so x_j depends only weakly on Q (Table 6). A hundred-fold increase in dose deepens the junction by only about 5.5 nm, whereas the peak concentration N_p rises by the full factor of 100.

Table 6: Calculated diode junction depth for the group's dose values ($R_p = 20\,\text{nm}$, $\Delta R_p = 5\,\text{nm}$, $N_B = 1 \times 10^{15}\,\text{cm}^{-3}$).

Q (cm^{-2})	1×10^{12}	3×10^{12}	1×10^{13}	3×10^{13}	1×10^{14}
N_p (cm^{-3})	7.98×10^{17}	2.39×10^{18}	7.98×10^{18}	2.39×10^{19}	7.98×10^{19}
x_j (nm)	38.3	39.7	41.2	42.5	43.8

2 Part B: Self-Aligned n-Channel MOSFET

2.1 Detailed Description of the MOSFET Process Flow

2.1.1 Layout

The layout file (`mos.cat`, top cell `Overall`) contains three mask layers, all centred on the origin (Table 7). The gate is a 30 nm long stripe running across the full width of the device in y. The two contact windows sit on either side of it. They are placed at $|x| \geq 50$ nm so that they clear the 30 nm nitride that later forms on the gate sidewalls (gate edge at $|x| = 15$ nm, nitride to $|x| = 45$ nm).

Table 7: Mask layers (coordinates in nm).

Layer	Rectangle(s) (x_1, y_1)–(x_2, y_2)	Purpose
`layer1`	$(-100, -100)$–$(100, 100)$	Simulation extent
`Gate`	$(-15, -100)$–$(15, 100)$	Polysilicon gate stripe ($L = 30$ nm)
`Contact`	$(-85, -30)$–$(-50, 30)$ and $(50, -30)$–$(85, 30)$	Source and drain contact windows

2.1.2 Process steps

The assignment's process flow was implemented as the 14 SEMulator3D steps listed in Table 8 and shown in Fig. 5.

Table 8: MOSFET process steps as implemented in SEMulator3D.

#	Assignment step	SEMulator3D action	Settings
1	Starting material	Wafer Setup	Si_Xtal, (100), 100 nm, boron 1×10^{15} cm^{-3}
2	Thermal oxidation	Straight Deposit	SiO_2 (thermal), 20 nm
3	Gate poly deposition	Straight Deposit	Si_Amorph (deposited Si used for poly), 50 nm
4		Planarizing Deposit	Photoresist, 50 nm above highest point
5	Gate patterning and etching	Expose Material	Mask `Gate`, polarity *Light* (resist kept on the stripe)
6		Straight Etch	Si_Amorph + SiO_2, selective, depth 70 nm, stops on Si
7		Lift-Off Wafer Op.	Strip photoresist
8	Ion implantation (S, D, poly)	Implant (Analytical)	As, 1×10^{15} cm^{-2}, $R_p = 20$ nm, $\Delta R_p = 5$ nm, radial range 0 nm; profiles in Si_Xtal and Si_Amorph; **no mask**
9	Nitride deposition	Conformal Deposit	Si_3N_4 (LPCVD), 30 nm
10	S/D contact patterning and etching	Planarizing Deposit	Photoresist, 50 nm above highest point
11		Expose Material	Mask `Contact`, polarity *Dark* (openings), 200 nm
12		Straight Etch	Si_3N_4, selective, depth 30 nm, stops on Si
13	Au metallization and lift-off	Straight Deposit	Au, 30 nm (directional)
14		Lift-Off Wafer Op.	Remove resist together with the Au on top of it

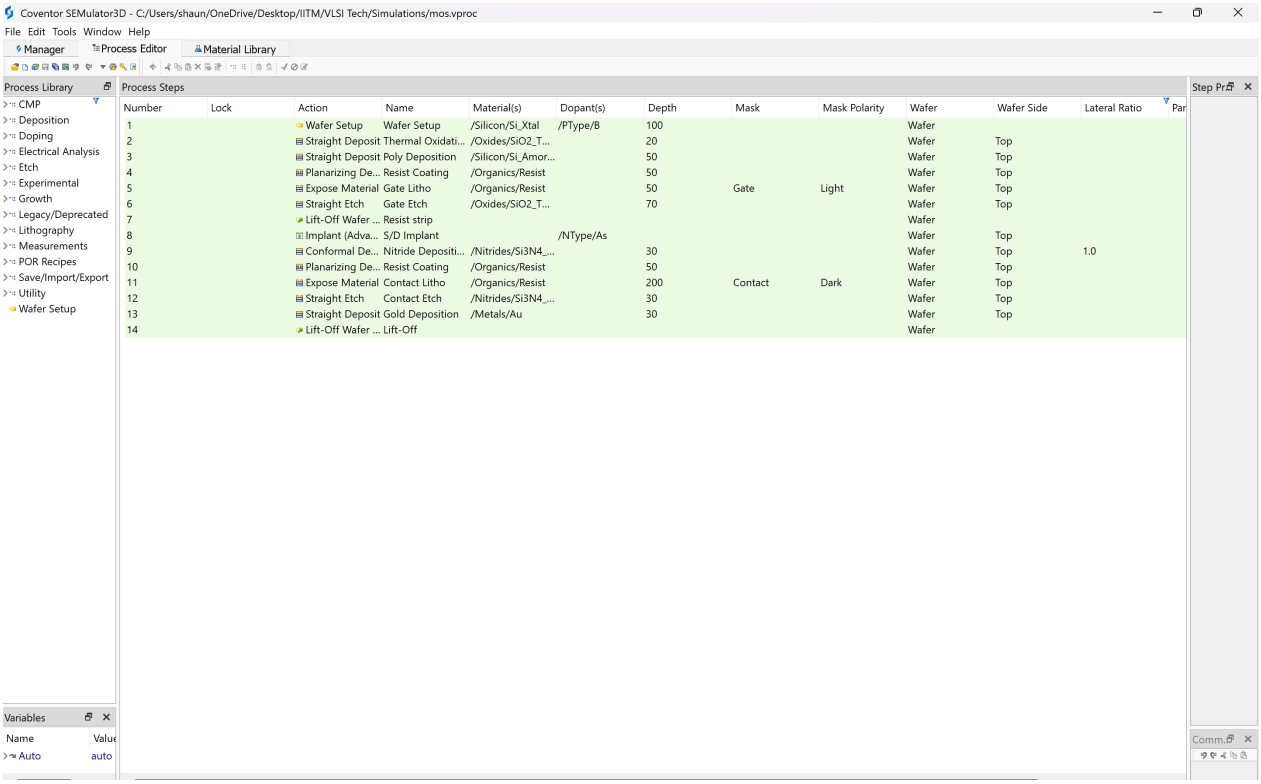


Figure 5: The 14-step MOSFET process in the SEMulator3D Process Editor.

2.1.3 Description of the flow

1. **Starting material.** A 100 nm p-type (100) silicon wafer, uniformly doped with boron at $1 \times 10^{15}\,\mathrm{cm}^{-3}$, forms the substrate and the channel region.

2. **Thermal oxidation.** A 20 nm thermal SiO_2 layer is grown over the whole wafer. Under the gate, this becomes the gate dielectric.

3. **Gate polysilicon deposition.** 50 nm of silicon is deposited for the gate electrode. The material database has no dedicated polysilicon entry, so the deposited (amorphous) silicon material `Si_Amorph` was used to represent the gate poly.

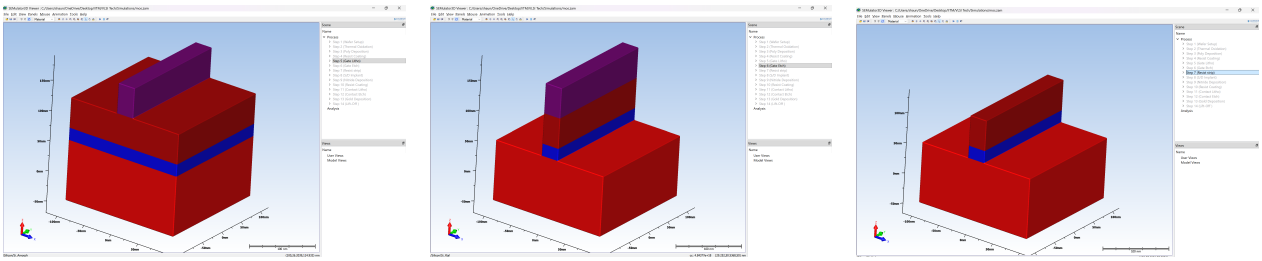
4. **Gate patterning and etching (steps 4–7).** Photoresist is applied and exposed through the `Gate` mask with *Light* polarity, so resist remains only on the gate stripe (Fig. 6a). A selective vertical etch then removes the exposed poly and oxide down to the silicon surface, and the resist is stripped. This leaves a 50 nm poly / 20 nm oxide gate stack on bare silicon (Fig. 6b,c).

5. **Ion implantation (step 8).** Arsenic is implanted vertically at a dose of $1 \times 10^{15}\,\mathrm{cm}^{-2}$ with $R_p = 20\,\mathrm{nm}$ and $\Delta R_p = 5\,\mathrm{nm}$ (Fig. 7). *No mask is used.* The poly gate itself stops the ions from reaching the channel, so the n^+ source and drain form only on either side of the gate, and their inner edges line up with the gate edges automatically. This is what makes the device **self-aligned**: no separate alignment step can misplace the source and drain relative to the gate. The same implant also dopes the poly gate n^+, making it a conductive electrode. Implant profiles were defined for both Si_Xtal (substrate) and Si_Amorph (gate). The dopant stops at the surface of the exposed oxide sidewalls. The 50 nm poly gate is thicker than $R_p + 4\Delta R_p = 40\,\mathrm{nm}$, so the arsenic is fully stopped within the gate and does not reach the gate oxide or the channel beneath it.

6. **Nitride deposition.** A 30 nm conformal LPCVD Si_3N_4 layer covers the whole structure, including the top and sidewalls of the gate. It passivates the device and insulates the gate from the source/drain metal.

7. **Source/drain contact patterning and etching (steps 10–12).** Photoresist is applied and exposed through the `Contact` mask with *Dark* polarity, which opens windows over the source and drain. The exposed nitride is etched selectively down to the n^+ silicon. The resist is left in place for the lift-off.

8. **Contact metallization and lift-off (steps 13–14).** 30 nm of gold is deposited directionally. It lands on the silicon at the bottom of the contact windows and on top of the resist everywhere else. Stripping the resist (lift-off) removes the gold on top of it, leaving gold only in the source and drain contacts (Fig. 8).

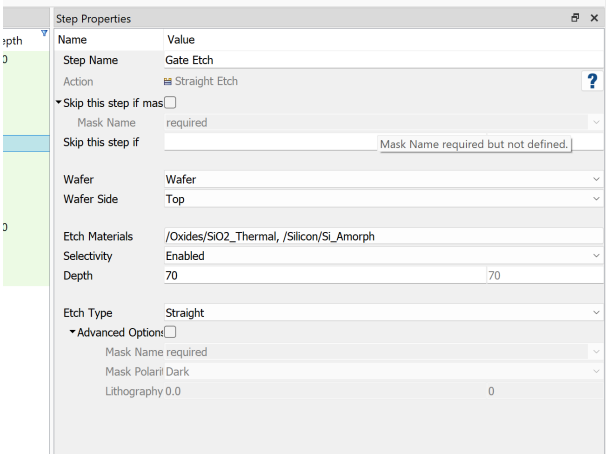


(a) Step 5: resist (purple) remains only on the gate stripe.

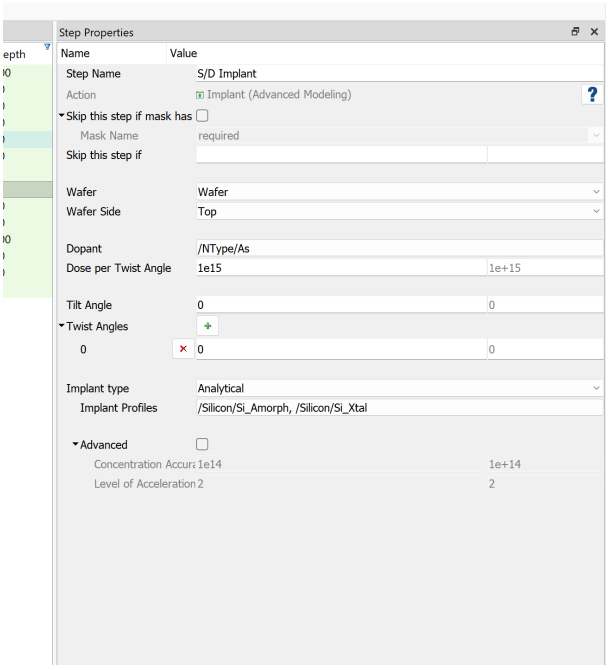
(b) Step 6: poly and oxide etched away outside the resist.

(c) Step 7: poly gate on gate oxide after the resist strip.

Figure 6: Gate patterning.



(a) Step 6: selective etch of Si_Amorph and SiO_2.



(b) Step 8: unmasked As implant into Si_Xtal and Si_Amorph.

Figure 7: Gate etch and self-aligned source/drain implant settings.

2.2 Final 3D Structure

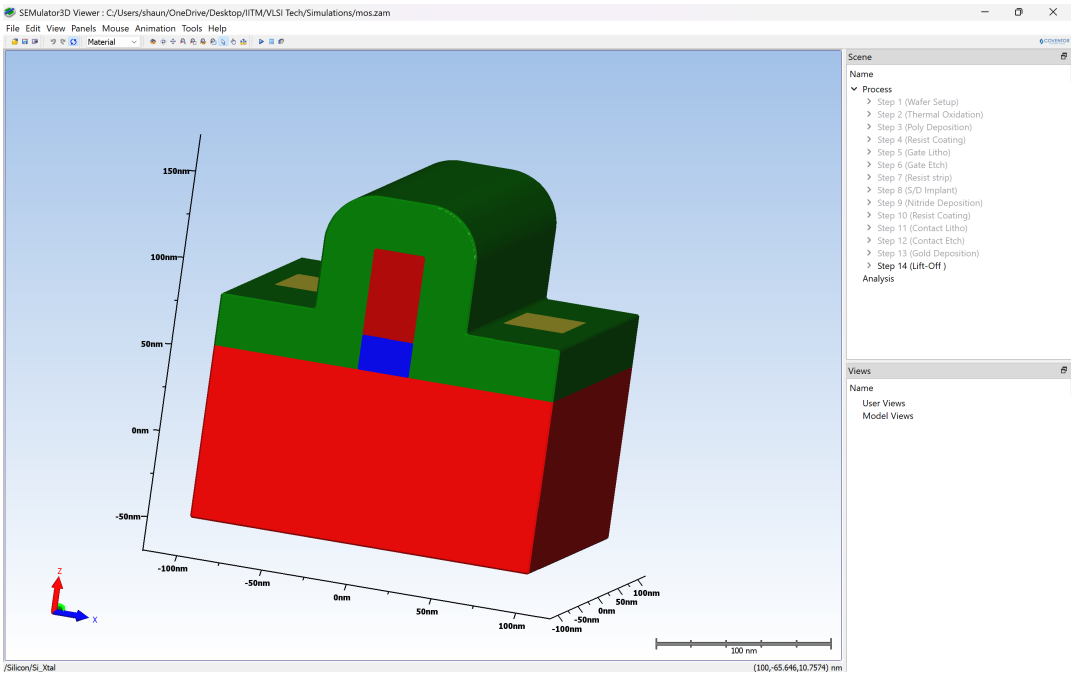


Figure 8: Final MOSFET structure after step 14. Red: p-Si substrate; blue: 20 nm gate oxide; dark red: poly gate; green: 30 nm Si_3N_4; gold: source and drain contacts at the bottom of the nitride openings on either side of the gate.

The build completed all 14 steps without errors (Fig. 9). The two warnings it reports are expected. In step 7, no floating material lay on the resist. In step 8, no implant profile was defined for SiO_2, so the dopant stops at the oxide surface.

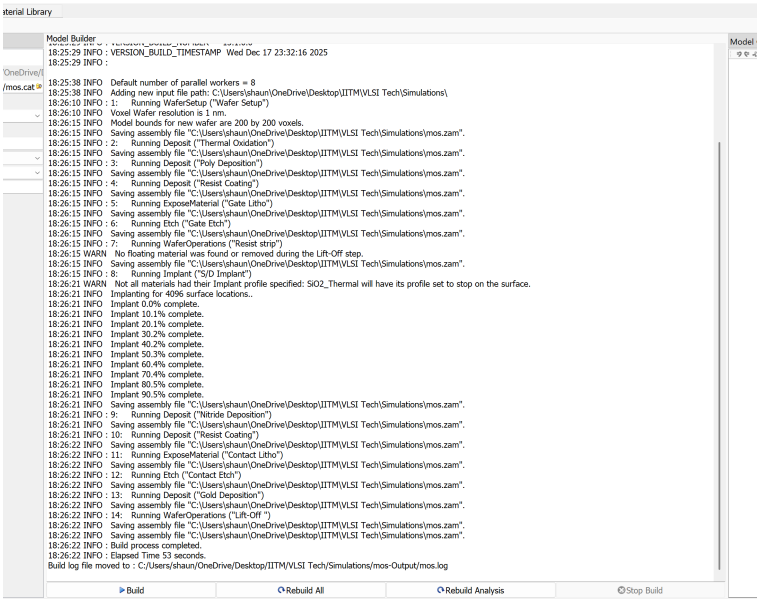


Figure 9: SEMulator3D build log for the MOSFET model.

2.3 Cross-Sectional View with Doping Profile

Figure 10 shows a cross-section through the centre of the device ($y = 0$, xz-plane) after the source/drain implant, coloured by the net doping $N_D - N_A$ (Total N–P). Two n^+ regions (source

and drain) sit on either side of the gate. No arsenic reaches the channel under the gate, and the poly gate is itself n^+ doped. This confirms the self-aligned implant.

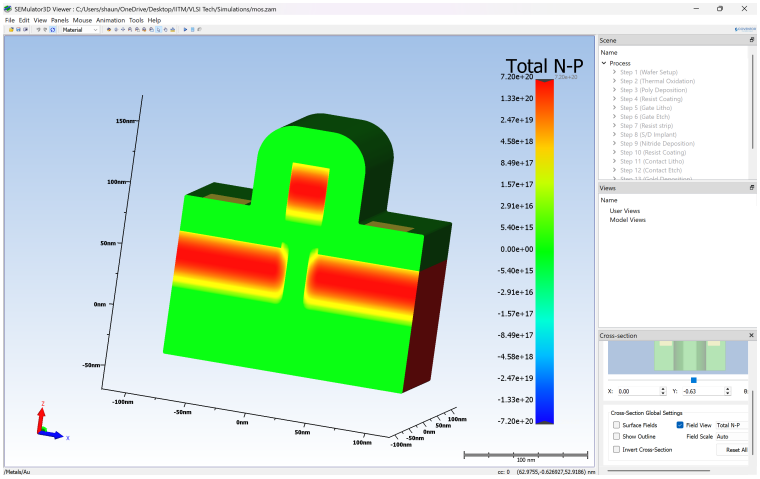


Figure 10: Cross-section of the MOSFET at $y = 0$ showing the net doping (Total N–P).

2.4 Calculation of Source/Drain Junction Depth

The arsenic was implanted into bare silicon (the oxide over the source and drain was removed in the gate etch), so the depth is measured from the silicon surface. Assuming an as-implanted Gaussian profile with no subsequent anneal,

$$N(x) = N_p \exp\left[-\frac{(x - R_p)^2}{2\Delta R_p^2}\right], \qquad N_p = \frac{Q}{\sqrt{2\pi}\,\Delta R_p}. \tag{1}$$

The metallurgical junction lies where the donor concentration equals the substrate acceptor concentration, $N(x_j) = N_B$, which gives

$$x_j = R_p + \Delta R_p \sqrt{2 \ln\left(\frac{N_p}{N_B}\right)}. \tag{2}$$

Values. $Q = 1 \times 10^{15}\,\mathrm{cm}^{-2}$, $R_p = 20\,\mathrm{nm}$, $\Delta R_p = 5\,\mathrm{nm} = 5 \times 10^{-7}\,\mathrm{cm}$, $N_B = 1 \times 10^{15}\,\mathrm{cm}^{-3}$.

Peak concentration.

$$N_p = \frac{1 \times 10^{15}}{2.5066 \times 5 \times 10^{-7}} = \frac{1 \times 10^{15}}{1.2533 \times 10^{-6}} \approx 7.98 \times 10^{20}\,\mathrm{cm}^{-3}.$$

Junction depth.

$$\frac{N_p}{N_B} = 7.98 \times 10^5, \qquad \ln(7.98 \times 10^5) = 13.590, \qquad \sqrt{2 \times 13.590} = 5.213,$$

$$x_j = R_p + 5 \times 5.213 = 20 + 26.07 \approx \boxed{46.1\,\mathrm{nm}} \quad (\text{for } R_p = 20\,\mathrm{nm}).$$

The second root of $N(x) = N_B$, at $R_p - 26.07\,\mathrm{nm} < 0$, lies above the silicon surface, so there is a single junction below the implant peak. Because Q, ΔR_p and N_B are common to the whole group, x_j increases one-for-one with the varied parameter R_p (Table 9).

Table 9: Calculated S/D junction depth for the group's range values.

R_p (nm)	10	15	20	25	30
x_j (nm)	36.1	41.1	46.1	51.1	56.1